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PROFESSIONAL SUMMARY:

- Generative AI Engineer with **10 years of experience** building and deploying LLM-powered solutions across **healthcare, insurance, banking, retail, and enterprise IT domains.**
- Specialized in **LLM applications, Retrieval-Augmented Generation (RAG), prompt engineering, and enterprise AI copilots** for knowledge retrieval and decision support systems.
- Strong expertise in designing **end-to-end GenAI pipelines** including document ingestion, embedding generation, vector search, LLM inference, and production deployment.
- Proven track record of delivering **high-impact GenAI solutions**, reducing processing time and improving decision efficiency across claims, fraud, and underwriting workflows.
- Hands-on experience with **OpenAI, Azure OpenAI, and Hugging Face Transformers**, building scalable applications for summarization, semantic search, and contextual Q&A.
- Extensive experience integrating **LLMs with structured and unstructured data** using vector databases and hybrid data architectures for enterprise use cases.
- Strong foundation in **machine learning and data science**, including fraud detection, risk modeling, anomaly detection, and recommendation systems.
- Expertise in **cloud-native GenAI deployments** using AWS and Azure with containerized architectures on Kubernetes for scalable and secure inference systems.
- Experienced in **GenAIOps and MLOps practices**, including CI/CD pipelines, MLflow, monitoring, and performance optimization of LLM-based applications.
- Skilled in implementing **secure and compliant AI systems**, including PII masking, access controls, and governance practices in regulated enterprise environments.
- Demonstrated leadership in **mentoring engineers, conducting code reviews, and driving best practices** in AI/GenAI system design and deployment.
- Strong experience in **real-time and batch inference systems**, enabling low-latency AI applications with scalable and resilient architectures.
- Adept at optimizing **LLM performance, cost, and latency** through prompt tuning, caching strategies, and efficient retrieval mechanisms.
- Collaborative engineer with experience working across **data engineering, DevOps, and business teams**, delivering production-ready, compliant, and scalable AI solutions.

TECHNICAL SKILLS:

Programming	Python, SQL, Node.js (JavaScript), PL/SQL, Bash/Shell Scripting
AI/ML	NLP, LLM Applications, RAG, Fraud Detection, Risk Modeling, Feature Engineering, Model Validation & Tuning
Frameworks	TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, Spark MLlib, Keras, spaCy, NLTK, Spring Framework (Spring Boot), Express.js, Symfony, Hibernate
Big Data	Apache Spark (PySpark), Hadoop (HDFS, Hive, HBase, Pig), YARN, Cloudera, Amazon EMR
Data Pipelines	Apache Airflow (1.x/), Apache Kafka, Apache NiFi, AWS Glue, Sqoop, ETL/ELT Design, Data Modeling, Snowflake Warehousing
Databases	Snowflake, Oracle 19c, SQL Server, MySQL, PostgreSQL, MongoDB, Neo4j (Graph DB), HDFS
Cloud	AWS (S3, IAM, Glue, Lambda, EKS, SageMaker), Azure (Azure OpenAI, Azure ML, Azure Data Factory, ADLS, Azure Databricks, AKS, Azure Cognitive Services), GCP (BigQuery, Cloud Storage, Vertex AI)
DevOps	Docker, Kubernetes (EKS, Red Hat OpenShift – On-Prem), Jenkins, TeamCity, Maven, CI/CD Pipelines, MLflow, SonarQube, ELK Stack, Prometheus, Grafana
Web & APIs	FastAPI, Flask, RESTful APIs, Spring Boot, Node.js, Express.js, React.js, Angular, Apache CXF
Tools	Git, GitHub, GitLab, Selenium, TestNG, Rally, JIRA, Confluence, Agile, Scrum, TDD, Code Reviews, Model Monitoring & Drift Detection, IAM & Security Best Practices

PROFESSIONAL EXPERIENCE:

New York Life Insurance - New York, NY **Oct 2024 to Present**
Generative AI Engineer

- Delivered real-time Generative AI solutions for claims intelligence and underwriting copilots, enabling AI-driven decision-making across insurance workflows.
- Solved challenges related to manual claims processing, fragmented policy analysis, and limited real-time decision support within underwriting systems.
- Generated 18% reduction in claims processing time and 15% improvement in underwriting turnaround using scalable AI architectures.
- Designed Retrieval-Augmented Generation (RAG) pipelines for document understanding, policy Q&A, and contextual reasoning using **Vertex AI, LangChain**, and **Hugging Face Transformers**.
- Built LLM-powered applications to automate claims document processing and policy analysis, reducing dependency on manual workflows.
- Connected structured data from **Amazon Redshift** and **Amazon RDS** with unstructured data in **Amazon S3** to enable context-aware retrieval and response generation.
- Engineered document ingestion, chunking, and embedding pipelines using **PySpark** and Python to support large-scale insurance datasets.
- Enhanced semantic search performance using **FAISS** and **Amazon OpenSearch (k-NN)** with hybrid retrieval and metadata filtering strategies.
- Developed API layers using **FastAPI** to serve LLM-based services for conversational AI and document query use cases.
- Deployed scalable GenAI services on **AWS EKS**, ensuring high availability, fault tolerance, and secure model inference.
- Applied parameter-efficient fine-tuning techniques **(LoRA/PEFT)** on open-source LLMs using GPU-based **AWS EC2** and **Vertex AI** pipelines.
- Automated end-to-end workflows using **Apache Airflow**, including document ingestion, embedding refresh, and vector index updates.
- Assessed system quality through evaluation pipelines measuring response accuracy, latency, and hallucination reduction using **Prometheus** and **Grafana.**
- Coordinated with data engineering, DevOps, and compliance teams to deliver scalable, secure, and production-ready AI systems.
- Enforced enterprise security standards through PII masking, **IAM-based access controls**, encryption using **AWS KMS**, and audit logging for LLM interactions.

Environment: Vertex AI, Amazon S3, Redshift, RDS, EC2, EKS, IAM, KMS, OpenSearch, Python, FastAPI, PySpark, FAISS, LangChain, LlamaIndex, MLflow, Airflow, Prometheus, Grafana, Docker, Kubernetes, Jenkins, Git/GitHub, SQL, Linux.

Citigroup - New York, NY **April 2022 to Sept 2024**
Generative AI/Data scientist

- Delivered scalable **AI/ML** and near real-time Generative AI solutions for financial document intelligence, fraud analytics, and enterprise knowledge retrieval systems.
- Addressed challenges in manual document review and delayed fraud investigations by implementing efficient **AI-driven solutions**.
- Improved document classification accuracy by 16% and reduced fraud false positives by 12% through advanced **machine learning models** and automation.
- Designed NLP and early-stage **LLM-based solutions** (Hugging Face Transformers) for summarization, entity extraction, regulatory Q&A, and transaction analysis.
- Transformed traditional pipelines into **RAG architectures (LangChain, FAISS)** to enhance contextual information retrieval.
- Integrated structured data from **Snowflake** with unstructured data in **Amazon S3** to support scalable feature engineering and retrieval workflows.
- Built batch and near real-time data pipelines using **PySpark** / **Apache Spark** and Hadoop (Hive) for large-scale processing.

- Optimized ETL workflows using **AWS Glue**, reducing pipeline latency and improving incremental data processing.
- Orchestrated end-to-end workflows using **Apache Airflow** for data ingestion, model inference, and retraining.
- Developed REST APIs using **Python (FastAPI)** to expose AI/ML services for fraud detection systems.
- Applied **Machine Learning (Scikit-learn, XGBoost)** for fraud detection, anomaly detection, and risk scoring.
- Leveraged graph analytics using **Neo4j** to model entity relationships and enhance fraud insights.
- Collaborated with cross-functional teams to deliver scalable and production-grade AI solutions.
- Implemented **CI/CD pipelines (Jenkins, Git)** to automate testing and deployment processes.
- Established validation and governance frameworks to ensure model accuracy, explainability, and regulatory compliance.

Environment: Python, Hugging Face Transformers, LangChain, FAISS, PySpark, Apache Spark, Hadoop (Hive), Snowflake, Apache Airflow, AWS (S3, EC2, IAM, Glue), Scikit-learn, XGBoost, Neo4j, FastAPI, Jenkins, Git/GitHub, SQL, Linux

The Home Depot - Atlanta, Georgia **Aug 2020 to March 2022**
AI/ML Engineer/Data Scientist

- Delivered real-time machine learning solutions for customer experience optimization, product recommendations, and intelligent search across eCommerce platforms.
- Addressed challenges related to low product discoverability, delayed customer insights, and lack of personalization in user interactions.
- Improved recommendation relevance by 14% and increased customer engagement by 12% through AI-driven personalization systems.
- Designed NLP models for sentiment analysis, product review classification, and text processing using **TensorFlow** and **Hugging Face (BERT).**
- Developed real-time recommendation systems using collaborative filtering and reinforcement learning techniques for personalized product suggestions.
- Connected streaming data from **Apache Kafka** and batch data from **Amazon S3** to enable continuous feature updates and real-time analytics.
- Engineered scalable data preprocessing and feature engineering pipelines using **PySpark** on **Amazon EMR** for large-scale retail datasets.
- Orchestrated batch and near real-time pipelines using **Apache Airflow**, enabling continuous ingestion, model updates, and workflow automation.
- Developed low-latency inference APIs using **Python** (**FastAPI**) to serve recommendation and search ranking results in real time.
- Applied machine learning models using **TensorFlow**, **PyTorch**, and **Spark MLlib** for recommendation, ranking, and prediction tasks.
- Implemented CI/CD pipelines using **Jenkins** along with model validation and monitoring frameworks to ensure performance, reliability, and production readiness.
- Collaborated with cross-functional teams including data engineering, product, and DevOps to deliver scalable, production-grade ML systems.

Environment: Python, TensorFlow, PyTorch, Hugging Face (BERT), Spark MLlib, PySpark, Apache Spark, Hadoop (HDFS), Apache Airflow, Apache Kafka, Snowflake, Amazon EMR, AWS (S3, EC2, IAM), Docker, Kubernetes, FastAPI, Jenkins, Git/GitHub, SQL, Linux

Anthem - Indianapolis, Indiana **Jan 2019 to July 2020**
Data Scientist / ML Engineer

- As a Data Scientist / ML Engineer at Anthem, developed machine learning solutions for healthcare claims automation and fraud detection using Azure-based data and analytics platforms.
- Aimed to reduce fraudulent claims, improve claims processing efficiency, and enable scalable analytics using cloud-native Azure services.
- Improved fraud detection accuracy by **25%** and reduced claims processing turnaround time by 30% through implementation of predictive models and automated scoring pipelines.

- Built and evaluated machine learning models including **Random Forest, XGBoost, Logistic Regression**, and deep learning models (TensorFlow/Keras) for fraud detection and anomaly identification.
- Designed and implemented scalable **data ingestion pipelines using Azure Data Factory and Kafka**, integrating on-prem Oracle data with Azure Data Lake Storage (ADLS).
- Engineered advanced **feature engineering pipelines using PySpark on Azure Databricks**, including aggregation, risk scoring, and provider-level behavioral analysis.
- Developed **NLP pipelines using spaCy and NLTK** to process medical records and claims documents for entity extraction and fraud signal identification.
- Managed model experimentation, tracking, and lifecycle using **Azure ML and MLflow**, ensuring reproducibility and efficient model management.
- Built and deployed **RESTful inference APIs using Flask/FastAPI**, containerized with Docker and deployed on Azure Kubernetes Service (AKS) for scalable inference.
- Implemented monitoring, validation, and compliance practices using **Azure Monitor**, **Application Insights, and enterprise security controls,** ensuring reliable and HIPAA-compliant ML systems.

Environment: Python, Machine Learning, NLP, TensorFlow, Keras, Scikit-learn, PySpark, Apache Spark, Kafka, Azure Data Factory, Azure Data Lake (ADLS), Azure Databricks, Azure ML, Azure Kubernetes Service (AKS), Azure Cognitive Services, SQL, REST APIs, Flask, FastAPI, Docker, MLflow, Git, Linux, Azure Monitor, Application Insights, CI/CD (Jenkins), Agile, Scrum.

GGK Technologies - Hyderabad, India **Aug 2016 to Dec 2018**
Software Developer

- Delivered backend-driven enterprise applications and data processing systems to support workflow automation and content management on on-premise infrastructure.
- Addressed challenges related to legacy system inefficiencies, scalability limitations, and fragmented business workflows across enterprise platforms.
- Improved application performance by 18% and reduced latency through backend optimization and efficient data handling.
- Designed scalable backend services using **Python** (**Django**, **Flask**) to implement business logic and workflow automation.
- Built RESTful APIs using **Django REST Framework** and Flask to enable integration across internal systems and third-party services.
- Modeled and optimized relational databases using **MySQL** and **Oracle (PL/SQL)** to support high-volume transactional systems.
- Migrated legacy applications from **Perl/PHP to Python**, refactoring monolithic systems into modular and maintainable architectures.
- Deployed and managed applications on **on-prem Linux servers (WebLogic, JBoss),** supporting environment configuration, release management, and production stability.

Environment: Python, Django, Flask, REST APIs, MySQL, Oracle (PL/SQL), XML, Linux, WebLogic, JBoss, Git, Agile/Scrum

Educational Details:

Sreenidhi University, Hyderabad
Bachelor of Technology